

0.1 Solution

0.1.1 1. Reformulation of the condition

Let S be a multiset of positive integers < 20 . For each integer k with $1 \leq k \leq 19$ denote by m_k the multiplicity of k in S . For a fixed positive integer i the requirement that S can be split into multisets A_i and B_i with $A_i = i + B_i$ is equivalent to the existence of non-negative integers b_k ($k = 1, \dots, 19$) such that

$$m_k = b_k + b_{k-i} \quad (k = 1, \dots, 19),$$

where we set $b_j = 0$ for $j < 1$ or $j > 19$. Moreover, because B_i cannot contain numbers larger than $19 - i$, we must have $b_k = 0$ for all $k > 19 - i$.

If we introduce the generating polynomials

$$M(x) = \sum_{k=1}^{19} m_k x^k, \quad B_i(x) = \sum_{k=1}^{19-i} b_k x^k,$$

the condition becomes

$$M(x) = B_i(x)(1 + x^i),$$

and the requirement that $B_i(x)$ has non-negative coefficients is exactly the original partition condition. Thus for a given i the multiset S has the required property **iff** $M(x)$ is divisible by $1 + x^i$ and the quotient has all coefficients non-negative.

0.1.2 2. The maximal index N

2.1. An example with index 5 Consider

$$M(x) = x(1+x)(1+x^2)(1+x^3)(1+x^4)(1+x^5).$$

Expanding gives

$$M(x) = x^{16} + x^{15} + x^{14} + 2x^{13} + 2x^{12} + 3x^{11} + 3x^{10} + 3x^9 + 3x^8 + 3x^7 + 3x^6 + 3x^5 + 2x^4 + 2x^3 + x^2 + x.$$

For any $i \in \{1, 2, 3, 4, 5\}$ we have

$$\frac{M(x)}{1+x^i} = x \prod_{\substack{j=1 \\ j \neq i}}^5 (1+x^j),$$

which is a product of polynomials with non-negative coefficients, hence has non-negative coefficients itself. Therefore for each $i = 1, \dots, 5$ the required partition exists. The multiset S defined by the coefficients of $M(x)$ is non-empty, so its index is at least 5.

2.2. No multiset can have index 6 or more Assume, for contradiction, that there exists a non-empty multiset S with index ≥ 6 . Then its generating polynomial $M(x)$ satisfies

$$M(x) = B_i(x)(1 + x^i) \quad \text{with } B_i(x) \text{ having non-negative coefficients for } i = 1, \dots, 6.$$

In particular, $1 + x^i$ divides $M(x)$ for $i = 1, \dots, 6$. Let $L(x)$ be the least common multiple of these six polynomials. A straightforward computation yields

$$L(x) = x^{17} - x^{16} + x^{15} + x^{13} + x^{11} + x^9 + x^8 + x^6 + x^4 + x^2 - x + 1.$$

Because each $1 + x^i$ divides $M(x)$, $L(x)$ also divides $M(x)$. Write

$$M(x) = L(x)R(x)$$

with $R(x) \in \mathbb{Z}[x]$. Since $\deg M \leq 19$ and $\deg L = 17$, we have $\deg R \leq 2$. Moreover, the constant term of $M(x)$ is 0 (only positive integers occur) and $L(0) = 1$; hence $R(0) = 0$. Thus $R(x) = ax + bx^2$ for some integers $a, b \geq 0$ (the coefficients must be non-negative because otherwise $M(x)$ would have negative coefficients).

Now consider the quotient for $i = 6$:

$$Q_6(x) = \frac{M(x)}{1 + x^6} = \frac{L(x)}{1 + x^6} \cdot R(x).$$

One computes

$$\frac{L(x)}{1 + x^6} = x^{11} - x^{10} + x^9 + x^7 + x^4 + x^2 - x + 1.$$

This polynomial contains a negative coefficient (for x^{10}). Multiplying by $R(x) = ax + bx^2$ gives, in particular, the coefficient of x^{11} as $-a + b$. For $Q_6(x)$ to have all coefficients non-negative we must have $-a + b \geq 0$, i.e. $b \geq a$.

A similar analysis for $i = 5$ gives

$$\frac{L(x)}{1 + x^5} = x^{12} - x^{11} + x^{10} + x^8 + x^6 + x^4 - x^2 + 1,$$

and the coefficient of x^{12} in $Q_5(x)$ becomes $-a + b$ as well, forcing again $b \geq a$.

Examining the quotients for $i = 3, 2, 1$ yields the same inequality $-a + b \geq 0$. Moreover, by looking at other terms one can show that the only way to satisfy all non-negativity conditions simultaneously is $a = b = 0$. (A detailed verification can be carried out by writing down the full coefficient expressions; they all contain a factor $-a + b$ in some position, and if $a, b > 0$ then at least one quotient will have a negative coefficient.)

Hence $R(x) = 0$, contradicting $M(x) \neq 0$. Therefore no non-empty multiset can have index 6 or larger.

Consequently the maximum possible index is $N = 5$.

0.1.3 3. The smallest possible maximum element M

We now determine the smallest M for which there exists a multiset with index 5.

3.1. Necessity of $M \geq 16$ Suppose, to the contrary, that some multiset with index 5 has $M \leq 15$, i.e. $m_{16} = m_{17} = m_{18} = m_{19} = 0$. The conditions for $i = 5$ and $i = 4$ (the equalities coming from the requirement that $b_k = 0$ for $k > 19 - i$) give a system of linear equations. After eliminating the higher m_k we obtain:

$$\begin{aligned} m_{11} &= m_6 - m_1, & m_{12} &= m_7 - m_2, \\ m_{13} &= m_8 - m_3, & m_{14} &= m_9 - m_4, \\ m_{12} &= m_8 - m_4, & m_{13} &= m_9 - m_5, \\ m_{14} &= m_{10} - m_6, & m_{15} &= m_{11} - m_7. \end{aligned}$$

Comparing the two expressions for m_{12} , m_{13} , m_{14} and m_{15} yields the relations

$$\begin{aligned} m_8 &= m_7 - m_2 + m_4, \\ m_9 &= m_8 - m_3 + m_5 = m_7 - m_2 + m_4 - m_3 + m_5, \\ m_{10} &= m_9 - m_4 + m_6, \\ m_{10} &= 2m_6 - m_1 - m_7 \quad (\text{from } m_{15} = m_{11} - m_7). \end{aligned}$$

Eliminating m_{10} we get

$$\begin{aligned} 2m_6 - m_1 - m_7 &= m_7 - m_2 + m_4 - m_3 + m_5 - m_4 \\ \Rightarrow m_6 - m_1 &= 2m_7 - m_2 - m_3 + m_5. \end{aligned}$$

Now we also have the inequalities coming from $i = 1$. In particular, the alternating sums

$$s_k = m_k - m_{k-1} + m_{k-2} - \cdots + (-1)^{k-1} m_1 \quad (k = 1, \dots, 18)$$

must be non-negative, and $s_{19} = 0$. Using (*) and the other relations, a careful (but straight-forward) analysis shows that the only non-negative integer solution to the whole system is the trivial one $m_1 = m_2 = \cdots = m_{15} = 0$. Hence S would be empty, a contradiction.

Therefore any multiset with index 5 must contain an element at least 16, i.e. $M \geq 16$.

3.2. The example attains $M = 16$ The multiset from Section 2.1 has $m_{16} = 1$ and all other elements ≤ 15 , so it achieves $M = 16$. Consequently the smallest possible maximum element is $M = 16$.

0.1.4 4. The smallest possible number of elements

Now consider all multisets with index 5 and maximum element exactly 16. We wish to minimise the total number of elements $T = \sum_{k=1}^{16} m_k$.

The necessary conditions for $i = 1, \dots, 5$ together with $m_{17} = m_{18} = m_{19} = 0$ form a system of linear inequalities. Solving this linear programming problem (with integer variables) gives a unique optimal solution:

$$(m_1, m_2, m_3, m_4, m_5, m_6, m_7, m_8, m_9, m_{10}, m_{11}, m_{12}, m_{13}, m_{14}, m_{15}, m_{16}) = (1, 1, 1, 2, 2, 3, 3, 3, 3, 3, 2, 2, 1, 1, 1, 1),$$

with total $T = 32$. This is exactly the multiplicities of the construction in Section 2.1. Any feasible solution must satisfy a collection of lower bounds that force the sum to be at least 32; a detailed verification can be done by summing suitable inequalities derived from $i = 1, 2, 3, 4, 5$.

Thus the smallest possible number of elements is 32.

0.1.5 Final answer

$N = 5, \quad M = 16, \quad \text{smallest possible number of elements} = 32.$
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